

Chinmoy Mondol

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Research Interests

My research centers on **computer vision**, **multimodal AI**, and **agentic AI**, with a growing interest in efficient perception for real-world systems. Recent work spans vision-transformer optimization for edge deployment, multimodal context understanding, and RAG-based conversational evaluation.

Education

BSc. in Computer Science and Engineering

American International University-Bangladesh (AIUB)

Sept 2017 – April 2021

Dhaka, Bangladesh

- CGPA: 3.99/4.00
- Thesis: "Music Suggestions from Determining the Atmosphere of Images" (Supervisor: Md Kishor Morol)

Research Experience

Research Assistant - Remote

ELITE Research Lab | Supervisor: Abir Bokhtiar

Sept 2026 – Present

Queens, NY, USA

- Collaborate with a 4-member research team to scope a novel research project in agentic AI and multi-agent systems, exploring intersections with computer vision and quantum computing.

Publications

- **Mondol, C.**, et al., "Real-Time Human Pose Estimation for Collaborative Robotics Using ROS 2 and Edge-Optimized Transformers," In preparation, target submission 2026.
- Nuhel, A. K., Taha, A. S., Ibrahim, M., Oishy, M. I., **Mondol, C.**, et al., "Integrating Visual Data in Conversational AI: A Study on LangChain Custom Tools for Image-Based Interaction," 2025 IEEE International Conference on Emerging Technologies and Applications (MPSec ICETA), Gwalior, India, 2025, pp. 1–6, doi: 10.1109/MPSecICETA64837.2025.11118796.
- Sohel, S. I., **Mondol, C.**, Ayon, H. S., Islam, U. T., & Morol, M. K., "Music Suggestions from Determining the Atmosphere of Images," 2021 24th International Conference on Computer and Information Technology (ICCIT), Dhaka, Bangladesh, 2021, pp. 1-7, doi: 10.1109/ICCIT54785.2021.9689781.

Research & Technical Skills

- **Research Skills:** Data Analysis, Literature Review, Academic Writing
- **Programming Languages:** Python, C, C++, Java
- **Machine Learning:** TensorFlow, Keras, scikit-learn, OpenCV, NumPy, Pandas, Jupyter Notebook, Kaggle
- **Databases:** SQL (MySQL, PostgreSQL), NoSQL (Redis)
- **Systems & Infrastructure:** Docker, Git
- **Writing Tools:** LaTeX

Awards & Honors

- Received a **Gold Medal** at the convocation ceremony in recognition of outstanding academic performance. 2021
- Graduated with the highest academic distinction, **Summa Cum Laude**. 2021
- Awarded **Dean's List Honors** four times during my tenure at AIUB. 2017 – 2021
- Awarded **Merit-Based Academic Scholarship (Tuition Fee Waiver)** from AIUB. 2017 – 2021
- Won **1st prize** in Inter-University ICT Olympiad at AIUB CS Fest. 2018

Teaching Experience

Teaching Assistant

American International University-Bangladesh (AIUB)

Fall 2021

Dhaka, Bangladesh

- Served as a TA in the "Artificial Intelligence and Expert System (CSC4121)" and "Web Technologies (CSC3222)" courses, assisting 12 lab sessions each and supporting 70+ undergraduate students across both.
- Designed programming assignments covering search strategies, adversarial search, and genetic algorithms (AI course), and tokenization and MVC architecture (Web Technologies)
- Assisted faculty members with grading assignments and final projects.

Professional Experience

Assistant Lead Engineer (QA)

Sept 2025 – Present

bKash Limited

Dhaka, Bangladesh

- Evaluate RAG-based conversational systems for a fintech assistant, building multi-dialect (Bangla/Banglish/English) test suites to stress-test retrieval and dialogue accuracy, and validating ASR/TTS pipelines for accent and financial-term precision.

Software Engineer (QA) → Senior Software Engineer (QA)

Jul 2021 – Sept 2025

Tiger IT Bangladesh Limited

Dhaka, Bangladesh

- Validated a real-time vehicle number plate recognition system, focusing on inter-service communication and Deep SORT integration.
- Collaborated with DevOps teams on containerized deployments, gaining hands-on exposure to Docker-based environments and deployment workflows.

References

Md Kishor Morol

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American International University-Bangladesh (AIUB)
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Dr. MD. Manzurul Hasan

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